

SQL Code

```
1  -- Descriptive Analysis:
2  /*
3  Q- What happened in the business?
4  Overall - what does overall business look like?
5  */
6
7  SELECT
8  SUM(net_sales) AS net_sales,
9  SUM(quantity) AS units_sold,
10 SUM(gross_profit) AS gross_profit,
11 ROUND(
12 SUM(gross_profit) / NULLIF(SUM(net_sales), 0) * 100,
13 2
14 ) AS gross_margin_pct,
15 SUM(discount_amount) AS discount_amount,
16 ROUND(
17 SUM(discount_amount)
18 / NULLIF(SUM(gross_list_value), 0) * 100,
19 2
20 ) AS discount_rate,
21 COUNT(*) AS sales_records,
22 COUNT(DISTINCT sku_id) AS active_skus,
23 COUNT(DISTINCT store_id) AS active_stores
24 FROM vw_sales_enriched;
25
26
27 -- Q How did sales performance change over time?
28 SELECT
29 DATE_TRUNC('month', date)::date AS month,
30 SUM(net_sales) AS net_sales,
31 SUM(quantity) AS units_sold,
32 SUM(gross_profit) AS gross_profit,
33 ROUND(
34 SUM(gross_profit)
35 / NULLIF(SUM(net_sales), 0) * 100,
```

```
36 2
37 ) AS gross_margin_pct,
38 COUNT(*) AS sales_records
39 FROM vw_sales_enriched
40 GROUP BY 1
41 ORDER BY 1;
42
43 -- Q Monthly Growth:
44 WITH monthly_sales AS (
45 SELECT
46 DATE_TRUNC('month', date)::date AS month,
47 SUM(net_sales) AS net_sales,
48 SUM(gross_profit) AS gross_profit,
49 SUM(quantity) AS units_sold
50 FROM vw_sales_enriched
51 GROUP BY 1
52 )
53 SELECT
54 month,
55 net_sales,
56 units_sold,
57 gross_profit,
58 ROUND(
59 (
60 net_sales
61 - LAG(net_sales) OVER (ORDER BY month)
62 )
63 / NULLIF(
64 LAG(net_sales) OVER (ORDER BY month),
65 0
66 ) * 100,
67 2
68 ) AS mom_sales_growth_pct
69 FROM monthly_sales
70 ORDER BY month;
71
72 -- Monthly Profitability:
```

```
73  SELECT
74  DATE_TRUNC('month', date)::date AS month,
75  SUM(net_sales) AS net_sales,
76  SUM(gross_profit) AS gross_profit,
77  ROUND(
78  SUM(gross_profit)
79  / NULLIF(SUM(net_sales), 0) * 100,
80  2
81  ) AS gross_margin_pct,
82  SUM(discount_amount) AS discount_amount,
83  ROUND(
84  SUM(discount_amount)
85  / NULLIF(SUM(gross_list_value), 0) * 100,
86  2
87  ) AS discount_rate
88  FROM vw_sales_enriched
89  GROUP BY 1
90  ORDER BY 1;
91
92  --# Store Performance Analysis:
93  SELECT
94  store_id,
95  store_name,
96  store_city,
97  store_type,
98  SUM(net_sales) AS net_sales,
99  SUM(quantity) AS units_sold,
100 SUM(gross_profit) AS gross_profit,
101 ROUND(
102 SUM(gross_profit)
103 / NULLIF(SUM(net_sales), 0) * 100,
104 2
105 ) AS gross_margin_pct,
106 SUM(discount_amount) AS discount_amount,
107 ROUND(
108 SUM(discount_amount)
109 / NULLIF(SUM(gross_list_value), 0) * 100,
```

```
110 2
111 ) AS discount_rate,
112 COUNT(*) AS sales_records
113 FROM vw_sales_enriched
114 GROUP BY
115 store_id,
116 store_name,
117 store_city,
118 store_type
119 ORDER BY net_sales DESC;
120
121 -- Category Performance:
122 SELECT
123 category,
124 SUM(net_sales) AS net_sales,
125 SUM(quantity) AS units_sold,
126 SUM(gross_profit) AS gross_profit,
127 ROUND(
128 SUM(gross_profit)
129 / NULLIF(SUM(net_sales), 0) * 100,
130 2
131 ) AS gross_margin_pct,
132 SUM(discount_amount) AS discount_amount,
133 ROUND(
134 SUM(discount_amount)
135 / NULLIF(SUM(gross_list_value), 0) * 100,
136 2
137 ) AS discount_rate,
138 COUNT(DISTINCT sku_id) AS sku_count
139 FROM vw_sales_enriched
140 GROUP BY category
141 ORDER BY net_sales DESC;
142
143
144 -- SKU & Brand Performance
145 SELECT
146 sku_id,
```

```
147     sku_name,
148     category,
149     subcategory,
150     brand,
151     SUM(net_sales) AS net_sales,
152     SUM(quantity) AS units_sold,
153     SUM(gross_profit) AS gross_profit,
154     ROUND(
155     SUM(gross_profit)
156     / NULLIF(SUM(net_sales), 0) * 100,
157     2
158     ) AS gross_margin_pct,
159     SUM(discount_amount) AS discount_amount
160 FROM vw_sales_enriched
161 GROUP BY
162     sku_id,
163     sku_name,
164     category,
165     subcategory,
166     brand
167 ORDER BY net_sales DESC
168 LIMIT 20;
169
170 -- 20 SKU performance:
171 SELECT
172     sku_id,
173     sku_name,
174     category,
175     brand,
176     SUM(net_sales) AS net_sales,
177     SUM(gross_profit) AS gross_profit,
178     ROUND(
179     SUM(gross_profit)
180     / NULLIF(SUM(net_sales), 0) * 100,
181     2
182     ) AS gross_margin_pct
183 FROM vw_sales_enriched
```

```
184 GROUP BY
185 sku_id,
186 sku_name,
187 category,
188 brand
189 ORDER BY gross_profit DESC
190 LIMIT 20;
191
192 -- Channel and Customer:
193 -- Which sales channels contribute most to sales, units, profit, and margin?
194 SELECT
195 channel,
196 SUM(net_sales) AS net_sales,
197 SUM(quantity) AS units_sold,
198 SUM(gross_profit) AS gross_profit,
199 ROUND(
200 SUM(gross_profit)
201 / NULLIF(SUM(net_sales), 0) * 100,
202 2
203 ) AS gross_margin_pct,
204 SUM(discount_amount) AS discount_amount,
205 ROUND(
206 SUM(discount_amount)
207 / NULLIF(SUM(gross_list_value), 0) * 100,
208 2
209 ) AS discount_rate,
210 COUNT(*) AS sales_records
211 FROM vw_sales_enriched
212 GROUP BY channel
213 ORDER BY net_sales DESC;
214
215 -- Loyalty Performance
216 --How do customer loyalty segments differ in commercial contribution?
217 SELECT
218 loyalty_segment,
219 COUNT(DISTINCT customer_id) AS customers,
220 SUM(net_sales) AS net_sales,
```

```
221     SUM(quantity) AS units_sold,
222     SUM(gross_profit) AS gross_profit,
223     ROUND(
224     SUM(gross_profit)
225     / NULLIF(SUM(net_sales), 0) * 100,
226     2
227     ) AS gross_margin_pct,
228     COUNT(*) AS sales_records
229 FROM vw_sales_enriched
230 WHERE customer_id IS NOT NULL
231 GROUP BY loyalty_segment
232 ORDER BY net_sales DESC;
233
234 -- Customer Value & Channel Preference
235 -- How does customer behavior differ across loyalty segments and actual sales channels?
236 SELECT
237     loyalty_segment,
238     channel,
239     COUNT(DISTINCT customer_id) AS customers,
240     SUM(net_sales) AS net_sales,
241     SUM(quantity) AS units_sold,
242     SUM(gross_profit) AS gross_profit,
243     ROUND(
244     SUM(net_sales)
245     / NULLIF(COUNT(DISTINCT customer_id), 0),
246     2
247     ) AS revenue_per_customer,
248     ROUND(
249     SUM(gross_profit)
250     / NULLIF(SUM(net_sales), 0) * 100,
251     2
252     ) AS gross_margin_pct
253 FROM vw_sales_enriched
254 WHERE customer_id IS NOT NULL
255 GROUP BY
256     loyalty_segment,
257     channel
```

```
258 ORDER BY
259 loyalty_segment,
260 net_sales DESC;
261
262
263 -- Discount & Inventory Position
264 -- How is sales activity distributed across discount levels?
265 SELECT
266 CASE
267 WHEN discount_pct = 0 THEN '0%'
268 WHEN discount_pct <= 5 THEN '1-5%'
269 WHEN discount_pct <= 10 THEN '6-10%'
270 WHEN discount_pct <= 15 THEN '11-15%'
271 WHEN discount_pct <= 20 THEN '16-20%'
272 WHEN discount_pct <= 25 THEN '21-25%'
273 WHEN discount_pct <= 30 THEN '26-30%'
274 ELSE '31%+'
275 END AS discount_band,
276 COUNT(*) AS sales_records,
277 SUM(quantity) AS units_sold,
278 SUM(net_sales) AS net_sales,
279 SUM(discount_amount) AS discount_amount,
280 SUM(gross_profit) AS gross_profit,
281 ROUND(
282 SUM(gross_profit)
283 / NULLIF(SUM(net_sales), 0) * 100,
284 2
285 ) AS gross_margin_pct
286 FROM vw_sales_enriched
287 GROUP BY 1
288 ORDER BY
289 MIN(discount_pct);
290
291
292 -- Inventory Position – 2025-10-31
293 -- What is the current inventory risk position across store-SKU combinations?
294 SELECT
```



```
295     inventory_status,
296     COUNT(*) AS store_sku_records,
297     SUM(stock_on_hand) AS total_stock_on_hand,
298     ROUND(
299     COUNT(*) * 100.0
300     / SUM(COUNT(*)) OVER (),
301     2
302     ) AS record_share_pct
303 FROM vw_inventory_risk
304 GROUP BY inventory_status
305 ORDER BY
306 CASE inventory_status
307 WHEN 'Below Safety Stock' THEN 1
308 WHEN 'Below Reorder Point' THEN 2
309 WHEN 'Normal' THEN 3
310 END;
311
312
313 -- Diagnostic Analysis:
314
315 -- Profitability Leakage:
316 -- Where are we generating sales but failing to convert them into strong profit?
317
318 SELECT
319     category,
320     SUM(net_sales) AS net_sales,
321     SUM(gross_profit) AS gross_profit,
322     ROUND(
323     SUM(gross_profit)
324     / NULLIF(SUM(net_sales), 0) * 100,
325     2
326     ) AS gross_margin_pct,
327     ROUND(
328     SUM(net_sales)
329     / SUM(SUM(net_sales)) OVER () * 100,
330     2
331     ) AS sales_contribution_pct,
```

```
332 ROUND(  
333 SUM(gross_profit)  
334 / SUM(SUM(gross_profit)) OVER () * 100,  
335 2  
336 ) AS profit_contribution_pct  
337 FROM vw_sales_enriched  
338 GROUP BY category  
339 ORDER BY net_sales DESC;  
340  
341  
342 -- Discount → Margin Driver Analysis:  
343 SELECT  
344 CASE  
345 WHEN discount_pct = 0 THEN '0%'  
346 WHEN discount_pct <= 5 THEN '1-5%'  
347 WHEN discount_pct <= 10 THEN '6-10%'  
348 WHEN discount_pct <= 15 THEN '11-15%'  
349 WHEN discount_pct <= 20 THEN '16-20%'  
350 WHEN discount_pct <= 25 THEN '21-25%'  
351 WHEN discount_pct <= 30 THEN '26-30%'  
352 ELSE '31%+'  
353 END AS discount_band,  
354 COUNT(*) AS sales_records,  
355 SUM(quantity) AS units_sold,  
356 SUM(net_sales) AS net_sales,  
357 SUM(discount_amount) AS discount_amount,  
358 SUM(gross_profit) AS gross_profit,  
359 ROUND(  
360 SUM(gross_profit)  
361 / NULLIF(SUM(net_sales), 0) * 100,  
362 2  
363 ) AS gross_margin_pct  
364 FROM vw_sales_enriched  
365 GROUP BY 1  
366 ORDER BY MIN(discount_pct);  
367  
368 --Product / Category Root Cause:
```

```
369 SELECT
370     category,
371     subcategory,
372     brand,
373     SUM(net_sales) AS net_sales,
374     SUM(gross_profit) AS gross_profit,
375     ROUND(
376     SUM(gross_profit)
377     / NULLIF(SUM(net_sales), 0) * 100,
378     2
379     ) AS gross_margin_pct,
380     ROUND(
381     SUM(discount_amount)
382     / NULLIF(SUM(gross_list_value), 0) * 100,
383     2
384     ) AS discount_rate
385 FROM vw_sales_enriched
386 WHERE category = 'Personal Care'
387 GROUP BY
388     category,
389     subcategory,
390     brand
391 ORDER BY gross_margin_pct;
392
393 -- SKU drivers:
394 SELECT
395     sku_id,
396     sku_name,
397     subcategory,
398     brand,
399     SUM(net_sales) AS net_sales,
400     SUM(gross_profit) AS gross_profit,
401     ROUND(
402     SUM(gross_profit)
403     / NULLIF(SUM(net_sales), 0) * 100,
404     2
405     ) AS gross_margin_pct,
```

```
406 ROUND(  
407 SUM(discount_amount)  
408 / NULLIF(SUM(gross_list_value), 0) * 100,  
409 2  
410 ) AS discount_rate  
411 FROM vw_sales_enriched  
412 WHERE category = 'Personal Care'  
413 GROUP BY  
414 sku_id,  
415 sku_name,  
416 subcategory,  
417 brand  
418 ORDER BY net_sales DESC;  
419  
420 -- Prescriptive Analysis:  
421 -- Which products/categories should be prioritized for discount review?  
422 SELECT  
423 category,  
424 subcategory,  
425 brand,  
426 sku_id,  
427 sku_name,  
428 SUM(net_sales) AS net_sales,  
429 SUM(gross_profit) AS gross_profit,  
430 ROUND(  
431 SUM(gross_profit)  
432 / NULLIF(SUM(net_sales), 0) * 100,  
433 2  
434 ) AS gross_margin_pct,  
435 ROUND(  
436 SUM(discount_amount)  
437 / NULLIF(SUM(gross_list_value), 0) * 100,  
438 2  
439 ) AS discount_rate  
440 FROM vw_sales_enriched  
441 WHERE discount_pct >= 21  
442 GROUP BY
```

```
443     category,
444     subcategory,
445     brand,
446     sku_id,
447     sku_name
448 ORDER BY net_sales DESC;
449 -- pricing/discount review based on actual commercial exposure--
450
451
452 -- Product Economics Actions:
453 SELECT
454     sku_id,
455     sku_name,
456     category,
457     subcategory,
458     brand,
459     MAX(list_price) AS list_price,
460     MAX(selling_price) AS realized_selling_price,
461     MAX(selling_price) -
462     MAX(list_price) AS price_gap,
463     MAX(selling_price) -
464     MAX(product_cost / NULLIF(quantity,0)) AS unit_profit,
465     SUM(net_sales) AS net_sales,
466     SUM(gross_profit) AS gross_profit,
467     ROUND(
468     SUM(gross_profit)
469     / NULLIF(SUM(net_sales),0) * 100,
470     2
471 ) AS gross_margin_pct
472 FROM vw_sales_enriched
473 WHERE category = 'Personal Care'
474 GROUP BY
475     sku_id,
476     sku_name,
477     category,
478     subcategory,
479     brand
```

480	HAVING SUM(net_sales) > 300000
481	ORDER BY gross_margin_pct ASC;